

Strategic Value Proposition & Cover Letter

Economic and Technical Case for Long-Term Contribution at Tumbleweed

Following my recent discussions with the Tumbleweed leadership team at EWOR Operators Day, I felt compelled to think beyond the immediate scope of the advertised position and instead evaluate a more fundamental question:

If Tumbleweed were to make the decision to sponsor my international relocation and work permit, does the long-term technical, operational and commercial value addition I can deliver significantly exceed the financial and administrative investment required to bring me onboard?

After conducting a preliminary systems-level assessment, I strongly believe the answer is unequivocally **yes**.

This document briefly outlines why.

I do not believe in merely performing assigned responsibilities.

I believe in **value addition work** — identifying opportunities where an existing system, technology, organisation or process can be strengthened, optimised and expanded in ways that create disproportionate long-term advantage.

That has consistently been the defining characteristic of my professional journey.

The Cost of International Hiring — A Quantifiable Investment

As an international candidate, I fully recognize that bringing me onboard requires additional commitment beyond a conventional local hire.

Estimated first-year employer-side costs (Netherlands):

One-Time Administrative Costs

- IND Recognised Sponsor Registration (<50 employees): **€2,539**
- Visa Application Fee: **€423**
- Legal / Documentation / Administrative Processing: **~€1,000**
- Relocation Assistance & Initial Transition Support: **~€2,500**

Estimated One-Time Cost: €6,000 – €7,000

Annual Employment Cost

- Minimum Highly Skilled Migrant Salary Requirement (Under 30): **€56,466 annually**
- Employer Taxes / Social Contributions / Administrative Overheads: **~€12,000**

Estimated First-Year Total Employer Commitment: ~€75,000

I view this not as a cost incurred for my benefit — but as an investment that should generate measurable return.

2. Engineering Value Addition Through Design Optimisation and Lean Manufacturing

One of the strongest immediate contributions I can make lies in improving engineering and manufacturing efficiency. Tumbleweed's mission architecture depends heavily on rapid payload integration, aggressive launch schedule, reliable structural design, fast prototyping and testing cycles and scalable manufacturing capability.

One immediate area I identified is implementation of **design optimisation workflows**, specifically:

- Design for Manufacturing (DfM)
- Design for Assembly (DfA)
- Parametric CAD optimisation
- Lean Engineering implementation
- Manufacturing process simplification
- Reduction of unnecessary structural complexity

This results in faster design validation, lower engineering rework, reduced manufacturing bottlenecks, reduced supply chain dependency and faster product iteration cycles. Engineering speed in hardware startups directly affects revenue velocity.

3. Additive Manufacturing Integration — Reducing Cost While Increasing Output

A second immediate area of value addition lies in transitioning critical structural manufacturing workflows toward advanced additive manufacturing technologies.

Specifically:

- Industrial-scale and production-ready machines by Markforged, EOS, Trumpf, 3D Systems, Lincsolution
- ESD-compliant and fire-resistant (FR) composite materials
- Carbon-fibre reinforced and metal additive manufacturing systems

These technologies can be strategically used for payload container structures, internal support structures, structural brackets, electronics housing assemblies, prototype structural hardware, rapid design validation cycles and even the spacecraft structure itself.

Advantages include:

- Lower material wastage
- Reduced CNC dependency
- Lower outsourced manufacturing cost
- Faster prototyping turnaround
- Greater manufacturing autonomy

Estimated manufacturing cost reduction: €10,000 – €32,000 annually

4. Faster Manufacturing Cycles = Higher Launch Frequency = Increased Revenue

This point is commercially critical. Launch aggregators such as **RIDESpace** increasingly create launch opportunities on relatively frequent cadence — often approximately every three months. This creates a simple operational reality.

If payload integration and manufacturing systems are slow: **Revenue opportunities are lost.**

To exploit higher launch cadence, engineering systems must support:

- faster prototyping,
- faster validation,
- faster payload readiness,
- reduced manufacturing delays,
- lean production management,

reducing prototype turnaround from **2–3 weeks to 48–72 hours** directly improves launch readiness. Despite the math being an oversimplification, it is logical - more launches create more revenue opportunities. This is not engineering optimisation; this is direct commercial advantage.

5. Product Lifecycle Improvement Through Radiation Engineering and Reliability Analysis

Companies operating within orbital manufacturing environments depend on one principle above all else: **Mission reliability.**

My technical background involving:

- Radiation exposure simulations
- Materials engineering
- Structural degradation analysis
- Environmental survivability modelling
- Aerospace lifecycle simulations

can contribute toward improving:

- Payload survivability
- Radiation hardening research
- Material selection optimisation
- Long-duration experiment reliability
- Improved mission lifecycle performance

For biomedical and semiconductor customers operating microgravity experiments: **Reliability is the product.**

Improving reliability directly improves customer confidence and repeat business.

6. Technology Transfer Agreements and Commercial Partnerships

One of my strongest professional instincts is identifying ecosystem-level opportunities. I naturally think beyond isolated engineering execution and focus on how technological organisations create strategic advantage through partnerships. Tumbleweed possesses strong long-term potential through **technology transfer agreements, commercial partnerships and institutional collaborations.**

I immediately identified opportunities through:

United Kingdom

- UK Space Agency and relevant space clusters (e.g. Space South Central, Space East, Scottish Space Network)
- National Microgravity Research Centre (Swansea University)
- EPSRC advanced manufacturing programmes
- Medical Research Council biomedical experimentation partnerships
- Farnborough International Airshow strategic market exposure

Europe and International

- Partnership for identification of launch opportunities and launch support with RIDESpace
- Technical University of Denmark (DTU)
- University College London & University of Southampton

- ESA technical collaboration pathways
- NASA collaboration opportunities
- Research-driven technology transfer agreements with established space institutions and aerospace companies
- Software transfer agreements improving the quality of workflows and product while opening the door for Kaizen-based self-development within the organisation

Strong partnerships creates revenue opportunities, institutional credibility, faster customer acquisition, research-backed product development and long-term commercial resilience. A single successful partnership can generate value exceeding international hiring costs many times over.

7. Portfolio Diversification — Building Financial Stability Through Multiple Revenue Streams

One of the greatest risks faced by frontier engineering startups is overdependence on a single revenue model. I believe one of my strongest capabilities lies in identifying opportunities for systematic value addition by introducing adjacent revenue streams while strengthening existing technological capability.

Potential diversification pathways include:

Mission-as-a-Service (MaaS) Consultancy where we can provide:

- Mission planning
- Payload integration consulting
- Systems engineering advisory
- Technical mission support services

for research institutions and commercial customers.

COTS Avionics Manufacturing

Development and manufacturing of:

- Flight avionics systems
- Electrical subsystems
- Modular CubeSat electronics
- Payload avionics packages

for emerging small satellite companies.

Technical Training and Industry Education

Developing advanced programmes for:

- Universities
- Engineering colleges
- Industry professionals
- Research institutions

covering:

- Space systems engineering
- Aerospace structures
- Satellite design
- Advanced manufacturing
- Orbital mission design

These create recurring revenue independent of primary mission cycles. Financial security through multiple revenue streams improves long-term survivability.

8. Engineering for Society — Building Commercial Success with Meaningful Impact

I strongly believe engineering organisations should think beyond immediate commercial objectives.

Portfolio diversification and strategic partnerships create opportunities to align with:

United Nations Sustainable Development Goals (UN SDGs) including:

- Healthcare innovation
- Advanced education
- Sustainable industrial development
- Scientific accessibility
- Global technological collaboration

This strengthens the organisation's Corporate Social Responsibility positioning, public sector collaboration, institutional credibility and long-term reputation. Engineering should not merely build products. It should improve society.

9. I Specialize in Value Addition Work

One realization that has consistently defined my professional journey is that I rarely operate within the narrow boundaries of a conventional job title. My strongest professional characteristic is not simply engineering execution - it is **value addition work**.

I naturally identify technical bottlenecks, operational inefficiencies, unrealised partnership opportunities, underutilised assets, manufacturing inefficiencies and ecosystem-level strategic opportunities; and instinctively focus on how systems can become stronger, faster and more commercially resilient.

Across previous roles, I have repeatedly contributed across engineering optimisation, startup operational development, product development strategy, technology partnerships, laboratory establishment, technical team hiring and development, international R&D policy and systems thinking and long-term commercial strategy. I do not simply perform assigned work. I improve systems I become part of.

Conclusion

Estimated First-Year Employer Investment ~ €75,000

Estimated One-Time Cost ~ €6,000 – €7,000

Net Cost ~ €81,000 - €82,000

Conservative Long-Term Value Addition

- Manufacturing Cost Reduction
- Faster Product Development Cycles
- Higher Launch Readiness
- Increased Revenue Through Faster Mission Turnaround
- Improved Product Reliability
- Technology Transfer Agreements
- Commercial Partnerships
- Portfolio Diversification Through New Revenue Streams
- Expansion Into UK and European Markets

Estimated Long-Term Value Addition: €250,000 – €300,000+

I do not define myself by a title. I define myself by a simple question.

How much stronger, faster, more resilient and commercially successful can a system become once I become part of it?

That has been the defining philosophy of my professional journey. If Tumbleweed chooses to make the investment required to bring me onboard internationally, I am confident I can ensure that decision becomes economically irrational to regret.

I seek systems where I can add value far beyond what is expected of me.

Sincere Regards,
Debjyoti Sengupta.